

LANVEOC POULMIC AB, France

WMO: 071090

Lat: 48.2794N

Lon: 4.4394W

Elev: 88

StdP: 100.28

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.1	1.0	-4.2	2.7	1.7	-2.6	3.1	2.8	14.9	9.2	13.2	9.7	3.9	90	0.594

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.8	25.7	18.9	23.7	18.0	21.9	17.2	19.7	24.3	18.7	22.1	18.0	20.5	3.8	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.3	20.9	17.5	12.7	19.9	16.9	12.2	19.0	56.6	24.6	53.4	22.2	50.9	20.6	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.0	10.5	9.4	-2.4	29.5	1.7	2.4	-3.6	31.3	-4.6	32.7	-5.6	34.0	-6.8	35.8		
(5)			-3.4	21.8	1.9	1.5	-4.8	22.8	-5.9	23.7	-6.9	24.5	-8.3	25.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.0	7.4	7.3	8.8	10.4	12.9	15.6	17.3	17.0	15.8	13.2	10.1	8.2	(6)
(7)		DBStd	4.37	2.70	2.63	2.57	2.42	2.42	2.43	2.17	1.98	2.35	2.46	2.42	2.86	(7)
(8)		HDD10.0	345	86	79	53	22	4	1	0	0	1	4	28	69	(8)
(9)		HDD18.3	2338	339	308	296	237	169	88	48	51	81	158	247	314	(9)
(10)		CDD10.0	1086	5	4	15	35	93	169	225	217	176	104	31	13	(10)
(11)		CDD18.3	38	0	0	0	0	1	7	14	9	6	1	0	0	(11)
(12)		CDH23.3	212	0	0	0	1	8	48	87	44	21	2	0	0	(12)
(13)		CDH26.7	36	0	0	0	0	1	10	14	9	1	0	0	0	(13)
(14)	Wind	WSAvg	4.7	5.3	5.1	5.0	4.6	4.5	4.3	4.1	4.1	4.4	5.0	5.4	(14)	
(15)	Precipitation	PrecAvg	1057	127	104	67	83	66	56	63	67	110	128	126	(15)	
(16)		PrecMax	1322	246	238	189	190	137	145	145	128	156	179	248	234	(16)
(17)		PrecMin	826	26	15	9	17	11	11	15	10	6	25	28	23	(17)
(18)		PrecStd	136	61	54	41	47	36	36	33	36	41	37	56	56	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	14.3	19.4	22.0	24.3	28.1	28.2	25.7	20.6	16.6	14.4	(19)	
MCWB			12.0	10.6	12.2	15.1	17.2	19.9	20.2	20.6	18.6	17.1	14.4	12.6	(20)	
2%		DB	12.4	12.5	15.5	18.6	21.1	24.4	25.6	24.1	23.2	18.7	15.3	13.5	(21)	
		MCWB	11.5	10.5	11.1	13.1	15.9	18.0	19.0	18.8	18.1	16.0	13.9	12.4	(22)	
5%		DB	11.8	11.7	13.3	16.1	18.9	22.0	23.5	21.7	21.2	17.6	14.5	12.9	(23)	
		MCWB	10.9	10.3	10.4	11.9	14.7	17.1	18.1	17.5	16.9	15.4	13.2	11.9	(24)	
10%		DB	11.3	11.1	12.1	14.1	17.0	20.0	21.3	20.2	19.5	16.7	13.7	12.3	(25)	
		MCWB	10.3	9.9	10.2	11.0	13.7	16.1	17.1	16.8	16.3	15.0	12.5	11.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.5	12.0	12.9	15.6	18.0	20.8	21.0	21.4	20.2	17.7	15.5	13.3	(27)
MCDWB			12.7	12.6	17.4	20.2	22.7	27.0	26.9	26.6	24.3	19.6	16.0	13.7	(28)	
2%		WB	11.9	11.4	11.9	13.6	16.3	18.7	19.6	19.3	18.7	16.8	14.3	12.7	(29)	
		MCDWB	12.3	12.0	13.7	17.4	20.3	23.0	24.6	22.6	21.5	18.0	14.9	13.2	(30)	
5%		WB	11.2	10.7	11.3	12.4	15.2	17.5	18.5	18.4	17.8	16.1	13.5	12.2	(31)	
		MCDWB	11.6	11.4	12.6	15.1	18.2	21.1	22.2	20.6	19.8	17.1	14.2	12.8	(32)	
10%		WB	10.6	10.1	10.6	11.6	14.1	16.4	17.6	17.7	17.0	15.4	12.7	11.5	(33)	
		MCDWB	11.1	10.9	11.9	13.5	16.2	19.1	20.3	19.3	18.6	16.4	13.6	12.3	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.5	5.0	5.9	6.7	6.9	7.0	6.8	6.9	7.1	5.8	4.9	4.5	(35)
MCDBR			4.0	5.4	7.8	10.1	10.4	10.7	11.1	9.1	9.3	6.5	5.3	4.2	(36)	
MCWBR			4.1	4.3	4.8	5.7	6.0	5.9	5.7	5.0	4.9	4.3	4.5	4.1	(37)	
5% WB		MCDBR	3.9	4.4	6.1	8.2	8.9	9.5	9.6	7.7	7.8	5.5	4.9	4.1	(38)	
		MCWBR	4.3	4.2	4.4	5.1	5.5	5.6	5.4	4.7	4.9	4.2	4.7	4.4	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.335	0.349	0.374	0.382	0.385	0.379	0.379	0.369	0.358	0.356	0.337	0.328	(40)	
taud		2.399	2.381	2.329	2.325	2.362	2.405	2.406	2.456	2.467	2.459	2.426	2.368	(41)		
Ebn at Noon		725	801	837	868	878	883	877	873	848	784	716	676	(42)		
Edh at Noon		69	87	106	117	118	113	112	102	92	79	66	63	(43)		
(44)	All-Sky Solar Radiation	RadAvg	0.98	1.80	2.99	4.68	5.54	5.99	5.71	4.89	3.91	2.29	1.23	0.82	(44)	
(45)		RadStd	0.07	0.18	0.31	0.39	0.48	0.63	0.30	0.39	0.28	0.17	0.09	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	+1.50	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page